

# Inverting Conditional Probability

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# Introduction

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- We know how to compute  $\Pr(A \mid B)$  when we know  $\Pr(A \wedge B)$  and  $\Pr(B)$ .
- But often what we're *given* is a set of conditional probabilities, and we need to work out a different conditional probability.
- Today we'll learn how to **invert** conditional probabilities.

## Associated Reading

*Odds and Ends*, Chapter 8

A burglary was committed, and fingerprints resembling mine were found at the scene. Let:

- $I$  = I am innocent
- $F$  = these fingerprints were found

From forensic science, we can estimate  $\Pr(F \mid I)$ : the chance that fingerprints like mine appear at a random crime scene where I'm innocent.

But what matters for criminal responsibility is  $\Pr(I \mid F)$ . These are very different things, and conflating them is a serious error.

## The Table Method

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## A Real-World Problem

- 3% of men have disease  $D$ .
- A man you know has no special risk factors.
- There is a test for  $D$ : among those with  $D$ , 90% test positive; among those without, only 5% test positive.

This man tested **positive**. What is the probability he has disease  $D$ ?

|                                                                     |
|---------------------------------------------------------------------|
| <b>condition positive (P)</b>                                       |
| the number of real positive cases in the data                       |
| <b>condition negative (N)</b>                                       |
| the number of real negative cases in the data                       |
| <b>true positive (TP)</b>                                           |
| exp. with hit                                                       |
| <b>true negative (TN)</b>                                           |
| exp. with correct rejection                                         |
| <b>false positive (FP)</b>                                          |
| exp. with false alarm, Type I error                                 |
| <b>false negative (FN)</b>                                          |
| exp. with miss, Type II error                                       |
| <b>sensitivity, recall, hit rate, or true positive rate (TPR)</b>   |
| $TPR = \frac{TP}{P} = \frac{TP}{TP + FN} = 1 - FNR$                 |
| <b>specificity, selectivity or true negative rate (TNR)</b>         |
| $TNR = \frac{TN}{N} = \frac{TN}{TN + FP} = 1 - FPR$                 |
| <b>precision or positive predictive value (PPV)</b>                 |
| $PPV = \frac{TP}{TP + FP} = 1 - FDR$                                |
| <b>negative predictive value (NPV)</b>                              |
| $NPV = \frac{TN}{TN + FN} = 1 - FOR$                                |
| <b>miss rate or false negative rate (FNR)</b>                       |
| $FNR = \frac{FN}{P} = \frac{FN}{FN + TP} = 1 - TPR$                 |
| <b>fail-out or false positive rate (FPR)</b>                        |
| $FPR = \frac{FP}{N} = \frac{FP}{FP + TN} = 1 - TNR$                 |
| <b>false discovery rate (FDR)</b>                                   |
| $FDR = \frac{FP}{FP + TP} = 1 - PPV$                                |
| <b>false omission rate (FOR)</b>                                    |
| $FOR = \frac{FN}{FN + TN} = 1 - NPV$                                |
| <b>Prevalence Threshold (PT)</b>                                    |
| $PT = \sqrt{\frac{TPR \cdot (TNR + 1) + TNR - 1}{(FPR + TNR - 1)}}$ |
| <b>Threat score (TS) or critical success index (CSI)</b>            |
| $TS = \frac{TP}{TP + FN + FP}$                                      |

Figure 1: Testing terminology (sensitivity and specificity)

## How to Solve These Problems

Build a table with:

- The possible **states of the world** on the rows (Has Disease / No Disease)
- The possible **test results** on the columns (Positive / Negative)
- Fill each cell with  $\Pr(\text{state} \wedge \text{result})$

The key formula used throughout:

$$\Pr(X \wedge Y) = \Pr(X) \times \Pr(Y \mid X)$$



Start with what we know:

- $\Pr(\text{Disease}) = 0.03$ , so  $\Pr(\text{No Disease}) = 0.97$
- $\Pr(\text{Positive} \mid \text{Disease}) = 0.9$ , so  $\Pr(\text{Negative} \mid \text{Disease}) = 0.1$
- $\Pr(\text{Positive} \mid \text{No Disease}) = 0.05$ , so  
 $\Pr(\text{Negative} \mid \text{No Disease}) = 0.95$

$$\Pr(\text{Disease} \wedge \text{Positive}) = 0.03 \times 0.9 = 0.027$$

|            | Positive | Negative |
|------------|----------|----------|
| Disease    | 0.027    |          |
| No Disease |          |          |

$$\Pr(\text{Disease} \wedge \text{Negative}) = 0.03 \times 0.1 = 0.003$$

|            | Positive | Negative |
|------------|----------|----------|
| Disease    | 0.027    | 0.003    |
| No Disease |          |          |

$$\Pr(\text{No Disease} \wedge \text{Positive}) = 0.97 \times 0.05 = 0.0485$$

|            | Positive | Negative |
|------------|----------|----------|
| Disease    | 0.027    | 0.003    |
| No Disease | 0.0485   |          |

$$\Pr(\text{No Disease} \wedge \text{Negative}) = 0.97 \times 0.95 = 0.9215$$

|            | Positive | Negative |
|------------|----------|----------|
| Disease    | 0.027    | 0.003    |
| No Disease | 0.0485   | 0.9215   |

## Checking the Table

|            | Positive | Negative |
|------------|----------|----------|
| Disease    | 0.027    | 0.003    |
| No Disease | 0.0485   | 0.9215   |

- Top row sums to 0.03  $\square$  (that's  $\Pr(\text{Disease})$ )
- Bottom row sums to 0.97  $\square$  (that's  $\Pr(\text{No Disease})$ )
- Everything sums to 1  $\square$

We want  $\Pr(\text{Disease} \mid \text{Positive})$ .

$$\Pr(\text{Disease} \mid \text{Positive}) = \frac{\Pr(\text{Disease} \wedge \text{Positive})}{\Pr(\text{Positive})}$$

- $\Pr(\text{Disease} \wedge \text{Positive}) = 0.027$
- $\Pr(\text{Positive}) = 0.027 + 0.0485 = 0.0755$

$$\Pr(\text{Disease} \mid \text{Positive}) = \frac{0.027}{0.0755} \approx 0.36$$

## The Surprising Upshot

- The test looked pretty reliable: 90% sensitivity, 95% specificity.
- But testing positive only raises the probability of disease from 3% to about **36%**.

Why? Because the disease is **rare**. Even though the test is fairly reliable, false positives outnumber true positives when the base rate is low.

This pattern is general. When testing for rare conditions, a positive result is worrying but usually doesn't mean you almost certainly have the disease.



## General Strategy

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## The General A/B Strategy

For any problem where you want  $\Pr(A \mid B)$  from information of the form  $\Pr(B \mid A)$ ,  $\Pr(A)$ , etc.:

1. Make a  $2 \times 2$  table with  $A, \neg A$  on rows and  $B, \neg B$  on columns.
2. Fill each cell using  $\Pr(X \wedge Y) = \Pr(X \mid Y) \Pr(Y)$ .
3. Add up rows to get  $\Pr(A)$ ,  $\Pr(\neg A)$ ; columns to get  $\Pr(B)$ ,  $\Pr(\neg B)$ .
4. Use the conditional probability formula to find the answer.

## Example

Given:

- $\Pr(A) = 0.6$
- $\Pr(B \mid A) = 0.3$
- $\Pr(B \mid \neg A) = 0.8$

Find  $\Pr(B)$  and  $\Pr(A \mid B)$ .

From negation:  $\Pr(\neg A) = 0.4$ ,  $\Pr(\neg B \mid A) = 0.7$ ,  
 $\Pr(\neg B \mid \neg A) = 0.2$ .

| <hr/>    |          |
|----------|----------|
| $B$      | $\neg B$ |
| <hr/>    |          |
| $A$      |          |
| $\neg A$ |          |
| <hr/>    |          |

$$\Pr(A \wedge B) = 0.3 \times 0.6 = 0.18$$

$$\Pr(A \wedge \neg B) = 0.7 \times 0.6 = 0.42$$

$$\Pr(\neg A \wedge B) = 0.8 \times 0.4 = 0.32$$

$$\Pr(\neg A \wedge \neg B) = 0.2 \times 0.4 = 0.08$$

|          | <hr/> |          |
|----------|-------|----------|
|          | $B$   | $\neg B$ |
| <hr/>    |       |          |
| $A$      | 0.18  | 0.42     |
| $\neg A$ | 0.32  | 0.08     |
| <hr/>    |       |          |

|          | $B$  | $\neg B$ |
|----------|------|----------|
| $A$      | 0.18 | 0.42     |
| $\neg A$ | 0.32 | 0.08     |

- $\Pr(B) = 0.18 + 0.32 = 0.5$

$$\Pr(A \mid B) = \frac{0.18}{0.5} = 0.36$$

Learning  $B$  drops the probability of  $A$  from 0.6 to 0.36.

## Probability Rules

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It helps to see the algebra behind the table method.

### The Negation Rule

$$\Pr(\neg A) = 1 - \Pr(A)$$

This holds because  $A$  and  $\neg A$  are exclusive and exhaustive.



# The Multiplication Rule

From the definition of conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(A \wedge B)}{\Pr(B)}$$

Multiply both sides by  $\Pr(B)$ :

$$\Pr(A \wedge B) = \Pr(A \mid B) \Pr(B)$$

This is what we use to fill in every cell of the table.

$B$  is equivalent to  $(A \wedge B) \vee (\neg A \wedge B)$ , and those are exclusive, so:

$$\Pr(B) = \Pr(B \mid A) \Pr(A) + \Pr(B \mid \neg A) \Pr(\neg A)$$

This is what lets us compute column totals from the cells.

Combining the multiplication rule and total probability:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B \mid A) \Pr(A) + \Pr(B \mid \neg A) \Pr(\neg A)}$$

This is **Bayes's Theorem**. It's very famous, but note that it's just a compact way of writing out what the table method does.

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)}$$

The table method and Bayes's theorem always give the same answer. Use whichever you find easier. For most exam problems, the table is clearer and less error-prone.

## Base Rates

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## A Famous Example

A cab was involved in a hit-and-run at night. Two cab companies operate in the city: Green (85% of cabs) and Blue (15%).

A witness identified the cab as Blue. The court tested the witness under similar conditions: they correctly identified the color of cabs 80% of the time.

**What is the probability the cab was Blue?**

## Setting Up the Table

Let  $\Pr(\text{Blue}) = 0.15$ ,  $\Pr(\text{Green}) = 0.85$ .

From the reliability data:

- $\Pr(\text{Looks Blue} \mid \text{Is Blue}) = 0.8$
- $\Pr(\text{Looks Blue} \mid \text{Is Green}) = 0.2$

|          | Looks Blue               | Looks Green              |
|----------|--------------------------|--------------------------|
| Is Blue  | $0.15 \times 0.8 = 0.12$ | $0.15 \times 0.2 = 0.03$ |
| Is Green | $0.85 \times 0.2 = 0.17$ | $0.85 \times 0.8 = 0.68$ |

$$\Pr(\text{Is Blue} \mid \text{Looks Blue}) = \frac{0.12}{0.12 + 0.17} = \frac{0.12}{0.29} \approx 0.41$$

Even though the witness is 80% accurate, it is still **more likely the cab was green**.



## The Lesson: Base Rates

- When one possibility is much more likely *a priori*, it takes strong evidence to overturn it.
- Here, Green cabs outnumber Blue 85:15. So even a fairly reliable witness can't overcome that.

We often conflate the **direction** of recent evidence with its **strength**. The witness supports “Blue” over “Green”, but the overall balance still favors Green.

## A Caution About This Example

The cab example assumes the witness's accuracy is stated in a particular way. It could be read differently:

If “80% accurate” means:

- $\Pr(\text{Correct} \mid \text{Looks Blue}) = 0.8$  (i.e., whenever they say “Blue”, they’re right 80% of the time)

Then the answer would simply be 0.8. English descriptions of accuracy data are often ambiguous. Always check exactly what you are being told.

- Base rates matter enormously.
- “This evidence supports  $H$ ” is different from “ $H$  is now probably true.”
- Use the table method to avoid being misled by salient recent evidence.

## Multiple Hypotheses

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## Worked Example: The Urn Problem

An urn contains 4 marbles, each either blue or green. The number of blue marbles is equally likely to be 0, 1, 2, 3, or 4. We draw 3 marbles **with replacement** and observe: **blue, green, blue**.

**Question:** What is the probability the urn contains exactly 1 blue marble?

Let  $H_k$  = the urn contains  $k$  blue marbles, for  $k = 0, 1, 2, 3, 4$ .

**Prior:**  $\Pr(H_k) = \frac{1}{5}$  for each  $k$ . We start with no reason to favour any particular composition.

## Computing the Likelihoods

Let  $E$  = the sequence blue, green, blue. Since draws are **independent** (with replacement), we multiply:

$$\Pr(E \mid H_k) = \frac{k}{4} \times \frac{4-k}{4} \times \frac{k}{4} = \frac{k^2(4-k)}{64}$$

| $H_k$ | $k^2(4-k)$ | $\Pr(E \mid H_k)$ |
|-------|------------|-------------------|
| $H_0$ | 0          | 0                 |
| $H_1$ | 3          | $3/64$            |
| $H_2$ | 8          | $8/64$            |
| $H_3$ | 9          | $9/64$            |
| $H_4$ | 0          | 0                 |

## Applying Bayes's Theorem

$$\Pr(H_1 | E) = \frac{\Pr(E | H_1) \Pr(H_1)}{\sum_{k=0}^4 \Pr(E | H_k) \Pr(H_k)}$$

Since all priors equal  $1/5$ , the  $1/5$  factors cancel and the denominator is just the sum of the likelihoods (times  $1/5$ ):

$$\Pr(E) = \frac{1}{5} \cdot \frac{0 + 3 + 8 + 9 + 0}{64} = \frac{1}{5} \cdot \frac{20}{64} = \frac{1}{16}$$

$$\Pr(H_1 | E) = \frac{(3/64) \times (1/5)}{1/16} = \frac{3}{320} \times 16 = \frac{3}{20}$$

## What the Evidence Tells Us

With a uniform prior, the posterior is **proportional to the likelihood**. The evidence does all the work:

| $H_k$ | Likelihood | Posterior |
|-------|------------|-----------|
| $H_0$ | 0          | 0         |
| $H_1$ | 3/64       | 3/20      |
| $H_2$ | 8/64       | 8/20      |
| $H_3$ | 9/64       | 9/20      |
| $H_4$ | 0          | 0         |

The most likely urn composition after observing blue-green-blue is  $H_3$  (3 blue marbles). Intuitively: we saw two blues and one green, which is consistent with a 3:1 ratio. The uniform prior means the evidence speaks for itself.



- We move from probability to **decisions**.
- How should we act when the outcomes of our choices depend on uncertain facts?
- We'll introduce expected utility as the key tool.